COURSE SYLLABUS

Linux Administration Essentials

COURSE NUMBER	DURATION	AUDIENCE LEVEL	FORMAT
LNX-001	1 day · 5 modules	Beginner	Instructor-Led / Self-Paced

Course syllabus

COURSE OVERVIEW

This intensive one-day course provides foundational skills for administering Linux systems through command-line interfaces. Students will gain hands-on experience with user management, security configurations, networking, and system resource optimization using Ubuntu Linux in a virtualized environment.

AT COURSE COMPLETION

After completing this course, you will be able to:

- Configure user accounts and manage permissions in Linux environments using command-line tools
- Implement security best practices including SSH key authentication and firewall configuration
- Establish and troubleshoot network connectivity on Ubuntu systems using netplan and diagnostic utilities
- Monitor and optimize system resources using process management and disk analysis tools
- Execute essential administrative tasks via command-line interface and automate routine workflows

PREREQUISITES

- Basic understanding of operating system concepts
- Familiarity with computer file systems and directories
- Ability to operate Windows desktop environment

COURSE OUTLINE

MODULE 1 Linux Fundamentals and User Management

90 min

This module introduces essential Linux command-line operations and user account administration. Students will learn to navigate the file system, manage user accounts, and control access permissions.

LESSONS

- **Lesson 1:** Introduction to Linux Command Line
- **Lesson 2:** Linux File System Navigation in Practice
- **Lesson 3:** User Account Creation and Management
- **Lesson 4:** Group Management and Membership
- **Lesson 5:** Understanding Linux Permission Models
- **Lesson 6:** Modifying Permissions with chmod
- **Lesson 7:** File Ownership with chown and chgrp

LAB EXERCISES

- Managing Users, Groups, and Permissions in Linux

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Navigate the Linux file system using command-line tools
- Create and manage user accounts and groups
- Configure file and directory permissions
- Modify file ownership for collaborative environments

MODULE 2 Linux Security Configuration

85 min

This module covers essential security practices for Linux systems, including SSH configuration, key-based authentication, and firewall management using ufw.

LESSONS

- **Lesson 1:** Linux Security Architecture Overview
- **Lesson 2:** SSH Service Architecture and Configuration
- **Lesson 3:** SSH Key-Based Authentication
- **Lesson 4:** Understanding Firewall Fundamentals
- **Lesson 5:** Implementing UFW Firewall Rules
- **Lesson 6:** Security Hardening Best Practices

LAB EXERCISES

- Configuring SSH and Firewall Security

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Configure SSH service for secure remote access
- Implement key-based authentication for passwordless login
- Manage firewall rules using Uncomplicated Firewall (ufw)
- Apply security best practices for system access control

MODULE 3 Network Configuration and Troubleshooting

80 min

This module teaches fundamental networking concepts for Linux systems, including IP configuration, DNS settings, and connectivity troubleshooting using command-line tools.

LESSONS

- **Lesson 1:** Linux Networking Architecture Overview
- **Lesson 2:** Network Interface Identification and Management
- **Lesson 3:** Netplan Configuration Syntax and Structure
- **Lesson 4:** DNS Resolution and systemd-resolved
- **Lesson 5:** Network Connectivity Troubleshooting Tools
- **Lesson 6:** Service Port Verification and Network Testing

LAB EXERCISES

Configuring Static IP and Troubleshooting Network Connectivity

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Configure static IP addresses using netplan
- Manage DNS resolver settings
- Diagnose network connectivity issues using diagnostic tools
- Verify network service status and port availability

MODULE 4 System Resource Monitoring and Optimization

80 min

This module focuses on monitoring system performance, identifying resource bottlenecks, and optimizing system resources through process and disk management.

LESSONS

- **Lesson 1:** Understanding System Resources and the Linux Process Model
- **Lesson 2:** Process Monitoring with top, htop, and ps Commands
- **Lesson 3:** CPU and Memory Utilization Analysis
- **Lesson 4:** Process Priority and Nice Values
- **Lesson 5:** Process Termination with Kill Signals
- **Lesson 6:** Disk Space Monitoring with df and du Commands
- **Lesson 7:** System Log Analysis Using journalctl
- **Lesson 8:** Performance Baseline Establishment and Trending

LAB EXERCISES

- Monitoring and Managing System Resources

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Monitor system resources using command-line utilities
- Identify and manage resource-intensive processes
- Analyze disk usage and optimize storage allocation
- Interpret system logs for performance troubleshooting

MODULE 5 Automation and Essential Administrative Tasks

85 min

This module covers task automation using cron, shell scripting basics, and essential administrative procedures that streamline daily Linux system management.

LESSONS

- **Lesson 1:** Introduction to Linux Automation
- **Lesson 2:** Understanding the Cron System
- **Lesson 3:** Shell Scripting Fundamentals
- **Lesson 4:** Control Structures: Making Scripts Intelligent
- **Lesson 5:** File Manipulation and System Maintenance Scripts
- **Lesson 6:** Bringing It All Together: Automation Best Practices

LAB EXERCISES

- Automating System Maintenance with Cron and Shell Scripts

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Schedule automated tasks using cron and crontab
- Create basic shell scripts for administrative tasks
- Implement system maintenance procedures
- Execute routine administrative workflows efficiently

BOOTCAMP MODE — CAPSTONE LABS

One integrative Capstone Lab is generated per course day. Each capstone presents a new scenario requiring students to independently apply the combined skills from that day's modules — no procedure, no hints. Students determine their own approach. Capstone Labs are the afternoon activity in Bootcamp delivery and replace the lecture portion of standard Instructor-Led Training.

DAY 1 CAPSTONE   Modules 1, 2, 3, 4, 5

SKILLS TO BE INTEGRATED

- Create user accounts with specific UID and home directory configurations
- Assign users to groups and verify group memberships
- Set file and directory permissions using symbolic and numeric notation
- Modify ownership of files and directories for collaborative access
- Configure SSH service for remote access
- Generate and deploy SSH key pairs for passwordless authentication
- Configure firewall rules using ufw to allow or deny specific services
- Assign static IP addresses using netplan configuration files
- Configure DNS resolver settings in systemd-resolved
- Test network connectivity using ping, traceroute, and netstat commands
- Identify resource-intensive processes using top and ps commands
- Terminate processes using kill and killall commands
- Monitor disk usage with df and du commands
- Analyze system logs using journalctl for troubleshooting
- Schedule automated tasks using cron and crontab
- Create and execute shell scripts for routine administrative tasks

Capstone content is embedded in the student manuals and instructor guides — one section per day, after the final module for that day. A standalone Capstone Slides (.pptx) file is also included in the Exclusive download package.